Anshul Patil

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Education

Indian Institute of Information Technology, Surat **Jun 2024 – May 2028**
Bachelor of Technology *Surat, Gujarat*
– Coursework: Operating Systems, Distributed Systems, DBMS, Algorithms

Experience

LFX Mentorship 2026, CNCF – OpenKruise **Sep 2026 – Nov 2026**
Software Developer Intern, Selected Contributor *Remote*
– Selected for CNCF's OpenKruise Agents, a Kubernetes-native platform for orchestrating AI agent sandboxes; implementing an end-to-end open-source signed-token authentication path in Go.
– Building an IdentityProvider that mints asymmetrically signed JWT access and ID tokens with key rotation, plus OIDC discovery and JWKS endpoints served by the sandbox-manager, removing the external identity-provider dependency for community deployments.
– Integrating ID token issuance with the existing `SecurityTokenRefreshReconciler` for pre-expiry refresh and implementing `PropagateSecurityToken` to deliver tokens into sandbox runtimes with refresh-on-update and lifecycle cleanup; Keycloak and Dex integration, Ginkgo end-to-end tests, and a design proposal.

Kubescape (CNCF) **Aug 2026 – Present**
Member, Open Source Contributor *Remote*
– Kubescape project member with 90+ merged Go PRs across kubevuln (the in-cluster vulnerability scanner), the CLI, opa-utils, and docs.
– Extended the SecurityException API with an `affected` status and action statements, package-scoped entries with subcomponents, and per-vulnerability expiry; added ECR credential resolution from the ambient AWS identity.
– Fixed vulnerability-pipeline correctness bugs: VEX generated on only 2 of 5 CVE-scan paths, external VEX overriding the scanner's own statements, relevancy keyed on package name without version, and scan deduplication keyed on tag instead of image.
– Hardened the scanner: closed an SSRF guard bypass via IPv4-in-IPv6 addresses, bounded registry `Retry-After` waits, fixed Syft goroutine races and nil-SBOM dereferences, and scoped CI permissions; wrote exception-precedence docs.

Google Summer of Code 2026, BRL-CAD **2026**
Software Developer Intern, Selected Contributor *Remote*
– Selected to improve reliability and reproducibility in the Manifold C++ geometry library (upstream of BRL-CAD); 25+ merged PRs across CI, numerics, bindings, and documentation.
– Built cross-platform determinism checks using fixed mesh datasets, canonical artifacts, and SHA256 comparison, and landed bitwise-exact trigonometric and non-trigonometric math so results agree across compilers and platforms.
– Added CI lanes now running on every PR: a non-blocking base-vs-head benchmark guard with structured JSON history, a memory-usage guard, weekly scheduled benchmarks, and a Linux Clang ASan/UBSan sanitizer lane.
– Shipped a 3MF importer with tests, an integer divide-by-zero fix in `RefineToTolerance`, C bindings that no longer mutated temporary copies, cone support in `Cylinder`, and versioned gh-pages docs deploys.

Techvisio Design **May 2025 – Aug 2025**
Software Developer Intern *Thane, India*
– Built a full-stack analytics dashboard using React, Django REST Framework, and SQL APIs processing 10K+ daily user events.
– Integrated AWS S3 pipelines for 300K+ monthly records; cut backend API latency 40% via SQL tuning and caching.

Research

Domain Generalization in Chest X-ray Classification | PyTorch, Lightning, timm, Hydra, MLflow **2026**
– Lead author on a controlled benchmark across 16 ImageNet-pretrained CNN backbones spanning 7 families on CheXpert, holding preprocessing, optimizer, and seed protocol fixed so backbone is the only variable.
– Built the 13-pathology multi-label pipeline with configurable uncertain-label policies, Hydra config composition, and MLflow tracking, then audited it and fixed result-invalidating defects: an unwired seed making three seeds train identically, NaN-poisoned macro-AUROC, pretrained weights falling back to random init, and a label table wrong in 18 of 42 cells.

Gujarati Legal Document Corpus for Bail Order Understanding | Python, Playwright, Tesseract, Docker, OCR **2026**
– Building a bail-order corpus from the eCourts Gujarat portal (CrPC 436–439, 33 districts): Playwright scraper handling modal-embedded, session-bound PDF URLs, resume-safe JSONL state, and an automatic CAPTCHA solver using a local Ollama vision model.
– Diagnosed Gujarati text-extraction failures via font auditing, separating valid ToUnicode maps from legacy LMG-Arun/TERAFONT subsets, and reverse-engineered the LMG2Shruti converter into a deterministic glyph-to-Unicode lookup table with OCR fallback; surveyed 10 Indian legal-NLP papers, none documenting a reproducible scraping layer.

Projects

Stratum | Python, FastAPI, React, TypeScript, PostgreSQL, LLM APIs **2026**
– Engineered a full-stack, multi-tenant developer platform on a persistent semantic codebase model unifying PR review, deploy coordination, drift tracking, and incident investigation; GitHub App auto-review, 230+ tests.
– Designed a multi-lens "swarm" LLM review pipeline running parallel security, performance, and logic passes for risk scores and exportable reports, plus semantic conflict detection for deploys and an incident investigator across 6 observability sources.

Achievements

- **Selected, Claude for Open Source Program**, Anthropic.
- **Selected, Amazon ML Summer School 2026:** Competitive applied ML program led by Amazon scientists.
- **Finalist, Flipkart GRiD 2026.**
- **3rd Place, Speed Coding:** Top participants at IIIT Surat out of 100+ competitors.

Technical Skills

Languages: Python, C++, Go, SQL, Bash, JavaScript/TypeScript
Machine Learning: PyTorch, Lightning, TensorFlow, Transformers, NumPy, Pandas, LLM APIs; evaluation, benchmarking, reproducibility
Backend and Data: FastAPI, Django REST Framework, REST APIs, PostgreSQL, Redis, Celery, caching, vector databases
Cloud and Infrastructure: Docker, Kubernetes (CRDs, controller-runtime, operators), AWS, GitHub Actions, CI/CD, Linux, MLflow
Security: JWT, OIDC, JWKS, key rotation, RSA/ECDSA, X.509